RAMDEOBABA UNIVERSITY

Department of Computer Science and Engineering

PROJECT SYNOPSIS

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SEEKVAULT

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1.Background and Motivation

Users today store large numbers of personal and academic files — notes, assignments, reports, certificates, project documents, and photographs or screenshots — across nested folders. As collections grow, locating a specific file becomes difficult, particularly when the user recalls what a file was about rather than its exact filename. For instance, notes on “Candidate Elimination” may be stored under a generic name such as ML_Unit3.pdf, and a relevant photograph may carry only a camera-generated filename, making both invisible to keyword-based search.

Semantic search overcomes this limitation by representing document content as numerical embeddings and retrieving files by contextual meaning rather than exact keyword match. However, applying AI to personal files raises privacy concerns: documents often contain phone numbers, financial details, identification numbers and other sensitive data, and sending such content to cloud-based AI services increases the risk of exposure.

SeekVault is proposed as a local-first, privacy-first multimodal semantic file search and organization system covering both documents and images. Privacy-first indexing means that SeekVault does not simply index every file it can find: the user first defines which folders the application is authorized to access, and only files inside those approved locations are processed. The system continuously monitors these authorized locations so that newly created, modified, moved, or deleted files are reflected in the local index. Before searchable representations are created, the system examines extracted text, OCR output, and visual content for potentially sensitive information. The original files are never modified; instead, SeekVault controls what representation is stored in its search index. Depending on the detected sensitivity, a file may be fully indexed, indexed after sensitive entities are replaced by placeholders, or represented only by non-content metadata. This design minimizes unnecessary exposure while preserving useful search capability.

2. Problem Statement

Existing file-search mechanisms rely primarily on filenames, keywords, folder structure and metadata, and therefore fail when a user remembers what a file contains or what an image looks like rather than its exact name. For example, a user may remember “the screenshot showing my Java error” or “the photo of the red car near a building” without remembering the filename. Conventional search cannot reliably connect such descriptions to files. AI-based semantic retrieval can address this limitation, but using external services for personal files can create data-control and privacy concerns. There is therefore a need for a system that can understand both textual and visual content, accept natural-language descriptions, operate only within user-authorized folders, monitor that authorization over time, detect potentially sensitive information before indexing, and apply different indexing treatment without

modifying the original files. The system should also maintain useful retrieval quality while identifying duplicate or near-duplicate files.

3. Objectives

- Develop a multimodal semantic search engine that retrieves documents and images by meaning rather than filenames or exact keywords, using cross-modal (text-to-image) retrieval for visual content.
- Implement fully local document and image processing with continuous folder monitoring, so raw personal files remain on the user's machine and the searchable index stays synchronized with changes inside authorized folders.
- Develop a privacy-aware indexing pipeline that first verifies folder authorization, then detects potentially sensitive information in both text and images, applies tiered treatment (full / redacted / metadata-only), and re-verifies authorization at query time before returning a result.
- Enable natural-language search across text documents and image files, returning semantically ranked results with similarity scores.
- Implement duplicate and version detection using file hashing and content similarity.
- Experimentally evaluate the trade-off between privacy protection and retrieval quality across indexing strategies for both textual and visual content.

4. Literature Review and Research Gap

Research Gap:

Traditional Windows Search:

- Mainly uses filenames, keywords, metadata, and indexed content.
- Gap: Does not combine semantic + multimodal search with privacy-aware indexing and file/version management.

Windows Semantic Search:

- Provides concept-based search for documents and images on supported Copilot+ PCs.
- Gap: SeekVault combines semantic retrieval with user-selected folders, sensitivity-based indexing, and duplicate/version detection.

Microsoft Recall:

- Searches locally stored PC activity snapshots using natural-language and visual/textual content.

- Gap: Recall focuses on activity snapshots, while SeekVault focuses on the user's actual authorized files, including documents, images, duplicates, and versions.

Microsoft 365 Copilot / Semantic Index:

- Uses lexical and semantic indexing with permission-based access to organizational data.
- Gap: SeekVault targets a local, personal file environment with full, redacted, or metadata-only indexing.

Cloud AI File Assistants:

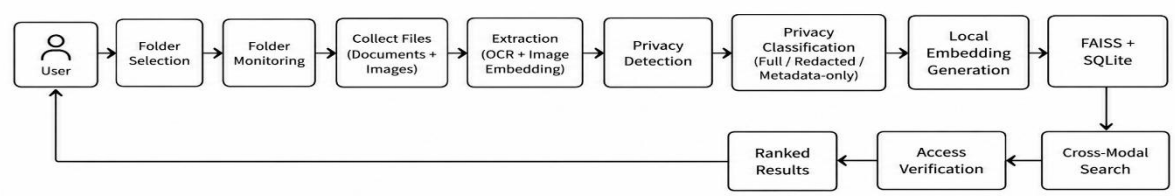
- Provide AI-based natural-language search and document retrieval.
- Gap: SeekVault follows a local-first architecture, keeping authorized files and searchable representations on the device.

OCR / Document Search Tools:

- Extract text from scanned documents and images.
- Gap: OCR alone does not combine visual + text semantic search, privacy-aware indexing, folder authorization, and file relationship management.

Sr.	Author(s) & Year	Title / Focus	Key Contribution	Limitation / Gap
1	Schumacher et al., 2008	Semantic desktop search	Fact + document retrieval via spreading activation	No local privacy-aware embedding pipeline
2	Minack et al., 2010	Beagle++ desktop search	Metadata + full-text indexing	Not privacy-aware semantic indexing
3	Reimers & Gurevych, 2019	Sentence-BERT	Efficient sentence embeddings for similarity	Text embedding technique only, no full app
4	Radford et al., 2021	CLIP cross-modal embeddings	Joint image-text embedding space for cross-modal retrieval	Cloud-scale training, not privacy-aware local search
5	Johnson et al., 2017	FAISS similarity search	Efficient billion-scale vector search	Retrieval efficiency only, not privacy-aware
6	Private-RAG, 2026	Privacy-preserving RAG	Privacy-preserving retrieval for LLM inference	Cloud-RAG focus, not local file search

5. Proposed Methodology / Plan of Work



6. Technology, Tools and Platforms

Frontend:

- Streamlit: Used to develop the user interface as a lightweight local web application. It allows users to select authorized folders, enter natural-language queries, view search results, and manage privacy settings without requiring a heavy desktop application.
- Pillow: Used for basic image loading and preprocessing before images are passed to the visual search pipeline.

Backend:

- Python: Acts as the core backend language and connects the file-processing, AI, privacy, search, and database modules.
- PyMuPDF & python-docx: Used to extract text and other relevant information from PDF and DOCX files.
- Tesseract OCR: Extracts text from scanned documents, screenshots, and text-containing images.
- Sentence Transformers: Converts extracted text and user queries into semantic embeddings, allowing searches based on meaning rather than exact keywords.
- CLIP/SigLIP: Generates image and text embeddings to support cross-modal searches, such as finding an image using a textual description.
- Microsoft Presidio + Custom Regex: Detects sensitive information such as emails, phone numbers, names, and other personal information before indexing.
- Watchdog: Continuously monitors authorized folders and detects new, modified, deleted, or moved files.
- Ollama (Optional): Provides local LLM/VLM capabilities for query understanding, image descriptions, and search-result explanations without sending data to external AI services.

Storage & Search:

- FAISS: Stores and searches the generated embeddings, enabling fast semantic and cross-modal retrieval.
- SQLite: Stores file metadata such as file path, type, timestamps, privacy classification, and indexing status.

7. Expected Outcomes, Deliverables and Functional Specifications

The expected result is a working local prototype in which privacy is enforced as part of the indexing and retrieval workflow rather than treated only as a statement about storage. A user first authorizes specific folders; SeekVault processes only those locations, continuously tracks changes within them, classifies the sensitivity of their content, and stores only the appropriate searchable representation locally. At query time, authorization is checked again before a result is returned. The system will support multimodal natural-language search across documents and images, sensitive-data detection, tiered

privacy-aware indexing, and duplicate/version grouping, with file contents remaining on the user's machine.

8. Project Scope

SeekVault is a local, single-user prototype designed for searching and organizing personal and academic files. Its scope includes:

- User-controlled access: Users select specific folders that SeekVault is allowed to access and index.
- Local file processing: Files within the approved folders are processed locally without sending their content to external servers.
- Multimodal files: Supports documents, images, screenshots, and scanned files.
- Semantic search: Enables natural-language search across both textual and visual content.
- Privacy-aware indexing: Files are classified as Full, Redacted, or Metadata-only based on detected sensitivity.
- Continuous monitoring: Tracks changes such as new, modified, deleted, or moved files within authorized folders.
- Access verification: Re-checks folder authorization before displaying a search result.
- File organization: Supports duplicate and version detection to help manage related files.

9. Project Timeline

The project is planned across an eight-week cycle. Early weeks establish architecture and core local-processing modules (folder access, extraction, embeddings, vector search); middle weeks add privacy detection, redaction and duplicate/version detection, the components of highest implementation complexity; the final week is reserved for testing, evaluation and documentation.

Phase	Duration	Milestone / Deliverable
Literature review & requirement analysis	Week 1	Finalized architecture and requirement document
Design & methodology finalisation	Week 1 – 2	Approved system design and module breakdown
Implementation/development	Week 2 – 7	Folder selection, extraction, embeddings, vector search, privacy detection, redaction, duplicate/version detection, UI integration
Testing & evaluation	Week 7 – 8	Precision@K/Recall@K/MRR results, privacy-detection F1, latency benchmarks
Documentation & final submission	Week 8	Final report, presentation and working demo

10. References

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